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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (CSE) (CBCS) Degree Examinations
May – June 2025

23CSE221 – WIRELESS NETWORKS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and one full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	What are the technical challenges of wireless communication? Explain multipath propagation, spectrum limitations, limited energy, and user mobility.	10	L*2	1	1
b)	Describe the structure and functionality of 802.11 generic data frames.	10	L2	2	1
2. a)	Illustrate the management frames in 802.11, focusing on SSID, TIM information elements.	10	L2	2	1
b)	Explain the MAC access modes. Compare and contrast the DCF and PCF.	10	L3	1	1
3. a)	Explain the Carrier-Sensing Functions and the Network Allocation Vector.	10	L2	1	1
b)	Describe the four types of control frames with neat diagram.	10	L2	2	1
Unit – II					
4. a)	Describe the following modulation techniques used in wireless networks, i) GFSK, ii) DPSK, and iii) Complementary Code Keying.	10	L2	3	1
b)	Explain the RF propagation in wireless communication?	10	L2	4	1
5. a)	Describe the 802.1x network port authentication.	8	L2	3	1
b)	Illustrate the different EAP methods used in 802.1x authentication?	8	L2	3	1
c)	Define spread spectrum and explain types of spread spectrum.	4	L2	4	1
6. a)	Illustrate the principles of Direct Sequence Spread Spectrum (DSSS) and its advantages in wireless communication.	10	L2	4	1
b)	Explain the concept of Robust Security Networks (RSN) in 802.11i.	10	L2	4	1
Unit – III					
7. a)	What is the cellular concept in wireless communication? Explain capacity enhancement and channel allocation techniques.	10	L2	5	1
b)	Explain the advancements of IPv6 in mobile wireless networks.	10	L2	5	1
8. a)	Compare and contrast traditional TCP with wireless TCP. What are the major challenges in implementing TCP in wireless domains?	10	L3	5	1
b)	Describe Wireless Application Protocol: WAP Model and protocol stack.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
